

SHAURYA SINGH

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PROFILE

MCA student and published researcher specializing in Deep Learning, Computer Vision, and Full-Stack Development. Builds end-to-end ML systems — from a peer-reviewed Devanagari OCR model to production-style face recognition and analytics applications — and evaluates them with the same rigor as a research paper: held-out test sets, ablation studies, and honestly reported metrics.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **ML / DL Frameworks:** PyTorch, Scikit-learn, DeepFace, torchvision
- **Web & Tools:** React, Git, Google Colab, Jupyter Notebook
- **Databases:** MySQL
- **Core Areas:** Deep Learning, Computer Vision, OCR, Data Analytics

PROJECTS

Devanagari (Marathi) Handwriting OCR — CRNN with Attention & CTC

2025 – Present

- Built an end-to-end CRNN pipeline (**ResNet-34 backbone + Squeeze-and-Excitation attention + 2-layer BiLSTM + CTC decoding**) for word-level Marathi OCR, achieving **95.4% character accuracy (0.0464 CER)** and **83.9% word accuracy** on a 15,000-image held-out test set
- Trained on **111,251 handwritten word images** (86,251 train / 10,000 val / 15,000 test, 117-character vocabulary) from the IIIT-Indic-HW-UC corpus, using targeted augmentation (rotation, blur, contrast/brightness jitter, shear) to improve robustness
- Designed and ran an **ablation study** isolating the SE-attention module, showing a consistent CER improvement (0.0464 vs. 0.0469) over an identical no-attention baseline
- Co-authored a research paper documenting the pipeline and benchmarking results against a published 20-language automotive HMI OCR verification system, prepared in Springer LNCs format

FaceOS — Production-Grade Face Recognition Platform

2025 – 2026

- Architected a fully containerized microservices platform (**Flask + Celery + Redis + PostgreSQL/pgvector + React 19**), orchestrated as 6 Docker Compose services, for face detection, recognition, and identity clustering at scale
- Replaced legacy VGG-Face embeddings with **ArcFace (512-dim) + RetinaFace detection**, and built a **pgvector IVFFlat ANN index** for $O(\log N)$ recognition lookups, using dual-threshold (L2 distance + match-margin) scoring to cut false positives
- Built an async processing pipeline (**Celery workers + Redis broker**) with live SSE progress streaming, SHA-256 upload deduplication, and custom IoU-based NMS to remove duplicate face detections
- Applied **DBSCAN clustering** to auto-group unidentified faces across a batch into distinct identities for bulk labeling, with a React 19 UI for tagging, correction, and live threshold tuning

Footlysis AI — Football Analytics Dashboard with NLP Query Engine

2025 – 2026

- Built an interactive analytics dashboard (**Panel/HoloViz + hvPlot/Bokeh**) covering 2,600+ players across the top 5 European leagues, with a custom glassmorphism UI and pan/zoom/hover-driven charts
- Built an **"Ask AI" natural-language query engine** using regex-based intent classification (top scorers, comparisons, stats, trends) and **fuzzy token-set matching (thefuzz)** to resolve player names from typos or partial queries, auto-syncing dashboard filters to the parsed result
- Designed a **hybrid data architecture** blending a static 2,600+ player dataset with live API-Football pulls for top scorers/assists across 5 leagues, backed by 24-hour local caching to stay within free-tier API limits while keeping load times fast

Against the Masters — Neural Network Chess Engine

2026

github.com/DA-Shaurya/against-the-masters

- Trained 10 champion-specific CNN policy networks (**17-plane board encoding, 4,096-move action space**) via imitation learning on grandmaster PGN data, then unified all champions into a single **dual-head (policy + value) network** with multi-task learning, using checkpointed, resumable training on Colab (T4 GPU)
- Implemented two independent chess engines client-side in JavaScript (**ONNX Runtime Web, WASM**): an **alpha-beta search with quiescence search and MVV-LVA move ordering** biased by the champion policy network, and a **PUCT-based Monte Carlo Tree Search** driven entirely by the learned value network in place of a hand-written evaluation function
- Found and fixed two critical engine bugs via targeted automated testing: a **game-state corruption bug** from unhandled exceptions during time-boxed search, and a **sign-inversion bug in MCTS node selection** that caused the engine to favor the opponent's best moves
- Deployed as a fully client-side, backend-free single-page app with drag-and-drop ONNX model loading, verifying **byte-for-byte parity** between the Python training pipeline's board encoding and the browser inference implementation

EDUCATION

Master of Computer Applications (MCA) — Pursuing

2025 – Present

MIT World Peace University (MIT-WPU), Kothrud, Pune

Bachelor of Computer Applications (BCA)

LPCPS, Gomti Nagar, Lucknow

EXTRACURRICULAR ACTIVITIES

- Actively solving Data Structures & Algorithms problems for placement preparation
- Exploring AI/ML research and full-stack/data analytics tooling beyond coursework